

HALLIBURTON FIELD, OK, USA

WMO: 722078

Lat: 34.471N

Lon: 97.960W

Elev: 339

StdP: 97.32

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53938

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.0	-6.1	-15.0	1.1	-5.5	-12.2	1.4	-2.6	12.1	6.6	11.2	7.3	3.7	10	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	39.7	23.6	37.4	23.7	35.9	23.7	25.9	32.4	25.4	32.4	25.0	32.1	4.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.8	28.1	23.7	19.3	27.8	22.9	18.4	27.2	82.1	32.9	79.9	32.0	77.6	32.3	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.6	8.6	-12.5	40.5	2.5	2.2	-14.3	42.0	-15.8	43.3	-17.2	44.5	-19.1	46.1
			-13.2	26.9	2.3	0.9	-14.8	27.6	-16.2	28.1	-17.5	28.6	-19.2	29.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	5.3	7.1	12.8	16.8	21.2	26.2	28.6	28.5	24.1	17.6	11.6	5.7	(6)
(7)		DBStd	9.46	5.27	6.05	5.58	4.57	4.25	2.39	2.80	3.18	3.44	4.73	5.02	5.28	(7)
(8)		HDD10.0	516	161	116	36	6	1	0	0	0	0	4	41	152	(8)
(9)		HDD18.3	1681	404	315	185	81	22	0	0	0	3	71	207	393	(9)
(10)		CDD10.0	3132	15	35	121	209	347	486	577	573	423	241	88	18	(10)
(11)		CDD18.3	1256	0	2	12	33	110	236	319	314	176	49	4	1	(11)
(12)		CDH23.3	14221	3	19	114	330	1023	2541	4061	3913	1719	447	47	3	(12)
(13)		CDH26.7	7154	0	3	21	86	376	1203	2320	2236	777	128	6	0	(13)
(14)	Wind	WSAvg	3.9	3.7	4.1	4.7	4.9	4.3	4.0	3.3	3.4	3.3	3.5	3.8	3.8	(14)
(15)	Precipitation	PrecAvg	900	39	45	70	92	119	106	64	74	84	90	60	52	(15)
(16)		PrecMax	1185	92	133	170	228	263	329	170	189	208	297	203	164	(16)
(17)		PrecMin	588	0	0	3	11	14	28	3	0	3	4	3	3	(17)
(18)		PrecStd	174	26	36	48	54	67	67	50	49	57	67	57	38	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.0	26.1	29.0	32.2	35.1	37.1	42.1	42.6	37.8	32.4	27.4	22.7	(19)
(20)		MCWB	12.8	15.3	18.2	19.2	22.3	24.0	23.2	23.6	22.2	21.9	18.2	14.1		(20)
(21)		2%	DB	20.1	22.5	26.3	28.9	32.5	35.2	39.9	40.0	34.1	29.9	24.3	19.1	(21)
(22)		MCWB	12.0	14.1	17.0	18.5	22.8	23.9	23.6	23.6	22.9	20.8	16.2	12.9		(22)
(23)		5%	DB	17.4	19.8	23.9	27.2	31.0	33.8	37.4	37.7	32.6	27.7	22.3	17.2	(23)
(24)		MCWB	11.2	13.5	15.9	18.3	22.2	23.8	23.8	23.8	22.7	19.6	15.6	12.0		(24)
(25)		10%	DB	14.7	17.2	22.1	25.0	28.3	32.5	36.0	36.0	31.3	26.1	19.9	14.1	(25)
(26)		MCWB	9.0	11.9	15.2	17.6	21.6	23.7	23.9	23.7	22.4	18.6	14.3	9.8		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	18.1	20.8	22.6	25.3	26.8	26.7	26.7	25.6	24.1	20.4	17.7	(27)
(28)		MDCB	19.2	21.5	24.3	27.9	31.5	32.2	32.8	32.8	31.7	28.1	24.3	20.2		(28)
(29)		2%	WB	14.3	16.3	19.3	21.4	24.2	25.6	25.8	25.8	24.7	23.0	18.4	15.5	(29)
(30)		MDCB	17.8	19.4	24.0	25.8	29.4	31.7	32.7	32.8	30.5	27.4	21.9	17.6		(30)
(31)		5%	WB	11.5	14.7	18.0	20.4	23.5	25.0	25.3	25.2	24.0	22.0	17.1	12.6	(31)
(32)		MDCB	16.1	18.6	21.4	24.2	28.2	31.2	32.8	32.9	29.4	25.6	19.6	15.2		(32)
(33)		10%	WB	9.2	12.2	16.7	19.3	22.7	24.3	24.9	24.7	23.4	20.6	15.7	10.4	(33)
(34)		MDCB	14.4	16.7	20.1	23.2	27.1	30.4	32.4	32.5	28.7	23.8	19.1	13.9		(34)
(35)	Mean Daily Temperature Range	MDBR	12.7	12.2	12.5	12.7	11.2	11.2	12.1	12.1	11.9	12.5	12.7	11.4	(35)	
(36)		5% DB	MCDBR	17.4	16.6	15.1	15.4	13.5	12.9	14.5	14.7	13.7	14.1	15.1	15.6	(36)
(37)		MCWBR	10.6	9.5	8.0	8.0	5.7	4.2	3.6	3.6	4.9	6.8	8.5	9.8	(37)	
(38)		5% WB	MCDBR	15.3	14.7	12.1	11.9	11.1	11.2	11.7	12.2	10.9	10.8	12.2	12.1	(38)
(39)		MCWBR	10.5	10.0	6.8	7.4	5.7	4.5	4.0	3.9	4.8	6.1	8.6	9.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.293	0.301	0.339	0.372	0.408	0.426	0.423	0.419	0.386	0.348	0.315	0.295	(40)	
(41)		taud	2.514	2.507	2.418	2.352	2.319	2.309	2.351	2.308	2.375	2.470	2.495	2.508	(41)	
(42)		Ebn at Noon	923	952	935	913	879	859	859	858	874	885	887	898	(42)	
(43)		Edh at Noon	82	92	110	123	129	130	124	127	114	95	83	78	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.90	3.48	4.61	5.65	6.04	6.78	6.85	6.26	5.20	3.99	3.07	2.48	(44)	
(45)		RadStd	0.30	0.51	0.29	0.46	0.64	0.62	0.40	0.36	0.46	0.56	0.39	0.28	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page